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Hardware-Aware TLS Identity Binding Profile

Draft v0.1.1

Abstract

This document defines a repository profile for binding upper-layer identity and authorization material to an accepted TLS 1.3 session and to post-handshake platform-attestation facts. The profile is application-protocol neutral at L0 through L2. Agent2Agent / AGTP is the first reference target for the upper layers, not a replacement for the trust model described here.

The core rule is simple: a verifier accepts a peer only when the peer's authenticated identity, session binding, freshness state, and local policy all refer to the same intended session, platform, service, agent, task, and authority boundary.

1. Status

This is a draft security-hardening profile for implementation review. It is not an IETF consensus document and it does not define a new TLS handshake, a new attestation evidence format, a new identity provider, or a new application core protocol.

Within this repository, this Markdown file is the normative source for the profile. Rendered specifications, notes, reports, and tests are derived from it unless they explicitly say otherwise.

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" are to be interpreted as described in BCP 14, RFC 2119 and RFC 8174 when, and only when, they appear in all capitals.

2. Terms

Hardware-aware TLS means ordinary TLS 1.3 plus post-handshake platform attestation and session binding. Platform attestation does not replace the TLS handshake and does not authenticate the platform before a TLS channel exists.

The older shorthand aTLS is reserved for existing Cocos implementation code paths, package names, or historical text. New profile text should use "hardware-aware TLS" unless it refers to that legacy surface.

Identity Grant means an authority statement issued by a Manager or another locally trusted policy authority. It authorizes upper-layer identity, task, and authorization values.

Session Binding Statement means a holder-of-key proof that binds one verified Identity Grant to one accepted TLS or exported-authenticator session. It does not authorize service, tenant, task, scope, resource, or capability values.

Expected values are verifier-local policy inputs. Observed values are values extracted from authenticated grants, session-bound statements, attestation claims, trusted manifests, or locally derived request state. Observed values do not become expected values without a trusted local policy decision.

3. Scope

This profile covers the checks needed before an application treats an accepted TLS peer as the intended platform, service, agent, task, or authorized actor. It focuses on fail-closed binding across three classes of facts:

1. accepted TLS and post-handshake attestation facts;
2. authenticated identity and authorization material;
3. verifier-local expected policy.

Out of scope:

- selecting a specific identity provider;
- standardizing a concrete wire-token format;
- changing attestation evidence formats;
- changing lower-layer TLS or post-handshake attestation wire protocols;
- proving a complete end-to-end authorization model;
- defining generic OAuth, OIDC, JWT, JWS, CWT, or COSE behavior except where this profile relies on them for fail-closed acceptance.

A requirement belongs in this profile when omitting it could cause any of these acceptance failures:

- a binding from the wrong live TLS or attestation session is accepted;
- the right platform is accepted for the wrong service, tenant, agent, task, or authority boundary;
- replay, stale state, missing state, or required-mode downgrade is accepted;
- peer-selected metadata becomes verifier policy;
- a security result, authorization decision, or verification result is reused outside the session or request for which it was produced.

4. Relationship to existing specifications

Most building blocks are already standardized. This profile composes them and adds repository-specific fail-closed requirements for identity, session binding, replay, canonical references, local policy comparison, and cache safety.

Area	Existing specification	Profile decision
Normative keywords OAuth roles and authorization vocabulary	BCP 14: RFC 2119 and RFC 8174 RFC 6749 and RFC 9396	Keywords apply to this repository profile only. Manager, Agent, relying party, scope, resource, and authorization_details are mapped by this profile.
JWT/JWS syntax and safety	RFC 7519, RFC 7515, and RFC 8725	The two-token Identity Grant plus Session Binding Statement model, agtp_type, agtp_version, grant_hash, and the mandatory claim set are profile choices.
Proof-of-possession pattern	RFC 7800 and RFC 8705	The Agent confirmation-key rule follows the PoP pattern. The exact Session Binding Statement claims are profile-specific.
TLS channel binding and exported authenticators	RFC 9266 and RFC 9261	The profile consumes accepted lower-layer binding facts. It does not define a new TLS exporter or exported-authenticator format.
Remote attestation roles	RFC 9334	The attestation-binder hash, accepted-evidence policy, and L1/L2 wiring are profile details.
CWT/COSE	RFC 8392 and RFC 9052	CWT/COSE is an alternative encoding for the same semantics, not a different trust model.

Area	Existing specification	Profile decision
HTTP response caching	RFC 9111	Response-cache vocabulary is reused. Security-state objects and verification results are not cacheable acceptance evidence.
OIDC	OpenID Connect Core	OIDC is vocabulary only. This profile does not require OIDC.
L0-L6, Identity Grant, Session Binding Statement, canonical references	Not standardized as one combined profile	These are local names for reviewable fail-closed identity binding.

5. Threat model

The attacker can observe, replay, reorder, or relay network messages. The attacker can provide malicious application metadata. The attacker can run another Agent on the same machine or in the same deployment environment.

In the Alice-side example, malware may influence peer-provided metadata or try to reuse old grants or binding statements. It does not fully control Alice's verifier process, local policy store, OS sandbox, keychain, or trusted Manager key configuration.

The lower layers are assumed to implement TLS 1.3 validation, exported-authenticator validation, and platform-evidence appraisal correctly. This profile defines the identity and policy material that must be bound to those lower-layer facts before the application accepts the peer as the intended Agent.

Out of scope for this threat model:

- a fully compromised verifier process or local policy store;
- a malicious or compromised trusted Manager or policy authority;
- compromise of all local secret storage, OS isolation, or keychain controls;
- denial of service;
- side-channel leakage outside the identity-binding path.

Threat	Required design consequence
Network relay or borrowed attestation	Bind the verified grant to the accepted TLS session, including endpoint-key hash, request-context hash, and attestation-binder hash when present.
Replay of old grants or bindings	Use iat, exp, unique IDs, and one-shot nonce or binding state. Required-mode deployments fail closed when replay state is unavailable.
Same-machine wrong-Agent	Keep Manager keys and Agent confirmation keys in separate trust domains. Accept only a Session Binding Statement signed by the grant-authorized confirmation key or by a key explicitly authorized by local policy.
Peer-provided metadata injection	Build accepted identity only from authenticated grants, session-bound statements, and local expected policy.

Threat	Required design consequence
Service, tenant, deployment, task, or capability diversion	Compare L3 through L6 values against local expected values. Required layers fail closed when expected values are missing or ambiguous.
Token substitution between grant and binding	Carry a domain-separated grant_hash over the exact signed grant bytes. Reject mismatched grant and binding pairs.
Key rotation, stale key, or revocation failure	Reject unknown, stale, retired, or revoked Manager keys, Agent binding keys, and grant IDs. Remote key sources need freshness and failure behavior.
Cache confusion or cross-authority leakage	Do not reuse identity tokens, binding statements, attestation evidence, authorization decisions, or verification results as acceptance evidence.
Gateway route confusion	In gateway-terminated deployments, authenticate the gateway-to-Agent route. Gateway session binding alone proves only the gateway endpoint.
Semantic alias or fuzzy-match confusion	Use canonical identifiers for decision-sensitive semantic values. Fuzzy matching, alias repair, or model interpretation is not an acceptance authority.

If Alice's verifier, local expected-policy source, or trusted key store is fully compromised, token design alone cannot recover security. Those cases require host isolation, policy-file integrity, secret isolation, process isolation, and operational controls outside this profile.

6. Layer model

The layer model separates lower-layer session and attestation binding from upper-layer deployment, agent, task, and authorization policy. Other profiles may collapse, rename, or extend these layers if they preserve the same fail-closed checks.

Layer	Verification target	Main failure class
L0	Live TLS channel	MITM or session confusion
L1	Attested platform validity	Fake, malformed, or untrusted platform evidence
L2	Attestation-to-channel or authenticator-to-channel binding	Relay, replay, or borrowed evidence
L3	Intended service, tenant, deployment, or environment	Service or tenant diversion
L4	Intended workload, process, or agent	Same-machine wrong-Agent
L5	Intended task, thread, context, or delegation	Cross-task or context diversion
L6	Intended authorization or capability policy	Confused deputy or privilege escalation

L0 through L2 can be tested with transport, attestation, and implementation regressions. L3 through L6 require explicit policy inputs before a verifier or application layer can enforce them consistently.

7. Application-protocol boundary

TLS and the hardware-aware profile are responsible for L0 through L2:

- authenticating the live TLS channel;
- appraising attested platform or VM evidence;
- binding authenticator or attestation material to the accepted TLS session.

An application profile carries or references the upper-layer identity and policy material needed for L3 through L6:

- intended service, tenant, deployment, or environment;
- intended workload, process, or Agent;
- intended task, thread, context, or delegation;
- authorization or capability policy.

Relay defense remains an L2 session-binding question. Diversion and wrong-Agent prevention require L3 and L4 policy comparison. Task and capability checks then continue at L5 and L6.

Current application-protocol drafts do not by themselves provide all semantic binding needed for service, tenant, Agent, task, capability, canonical-reference, replay, and local-policy checks. The application protocol may carry authenticated policy inputs, such as an Identity Grant, and bind them to the accepted TLS session through a Session Binding Statement. The verifier still compares the resulting observed values with local expected values before accepting the peer.

Review item	Question	Profile requirement	Negative-test anchor
SP-01	Is identity material bound to the accepted TLS and attestation session?	Verify the Session Binding Statement against endpoint key, request/exporter context, attestation binder when present, and replay state.	hwtls-id-profile-relay-001; attestation-binder mismatch
SP-02	Which values are local policy and which are peer claims?	Compare service, tenant, deployment, agent, task, scope, resource, and authorization values against local expected policy.	hwtls-id-profile-diversion-001; hwtls-id-profile-wrong-agent-001; hwtls-id-profile-binding-confusion-001
SP-03	Which freshness values are one-shot?	Reject repeated Session Binding Statement IDs, nonces, or task-binding values when one-shot use is required.	hwtls-id-profile-replay-001
SP-04	What happens when grant or binding material is missing, unsupported, substituted, or partly verified?	Required mode fails closed. Neither Identity Grant nor Session Binding Statement is sufficient alone.	hwtls-id-profile-downgrade-001; nested-token substitution

Review item	Question	Profile requirement	Negative-test anchor
SP-05	Which responses are safe for shared caching?	Treat caller-dependent responses as private with adequate partitioning or no-store. Do not cache security inputs or verification results as acceptance evidence.	caller-dependent cache leak
SP-06	Are semantic references canonical?	Reject non-canonical, ambiguous, fuzzy-matched, or receiver-repaired decision values.	semantic alias confusion
SP-07	Are Manager and Agent keys separated?	Reject Manager keys used as Agent confirmation keys and Agent keys used as Manager signing keys.	key-role confusion
SP-08	Is gateway routing authenticated?	In gateway mode, bind the gateway endpoint and the gateway-to-Agent route separately.	gateway route confusion

8. Authority and key separation

One implementation profile used in this repository is:

Identity Grant + hardware-aware TLS + OAuth/OIDC-style semantics + JWT/JWS encoding

OAuth and OIDC provide claim semantics and review vocabulary. JWT/JWS provides one signed-token encoding. CWT/COSE provides a compact binary encoding for the same trust model.

The profile keeps three key roles separate:

Key role	Purpose
TLS endpoint key	Proves possession for the accepted TLS or exported-authenticator endpoint.
Agent binding key	Signs the Session Binding Statement that binds a verified grant to the accepted TLS session.
Manager or policy-authority key	Signs Identity Grants that authorize intended deployment, Agent, task, or capability values.

These keys may be related by deployment policy, but they MUST NOT be treated as the same key by default. The Manager key is a token-signing or policy-authority key, not a TLS endpoint key.

Manager signing keys MUST NOT be accepted as Agent confirmation keys. Agent confirmation keys MUST NOT be accepted as Manager or policy-authority signing keys. A deployment MAY explicitly authorize an endpoint key for session binding, but that authorization must come from the verified grant or from local policy; it must not be inferred from the TLS session alone.

The JWS protected header kid is only a key-selection hint. Acceptance also requires issuer, audience, algorithm allow-list, key status, key use, profile version, token type, time validity, and local policy checks. The signing-method allow-list MUST NOT include none.

9. Identity Grant

An Identity Grant authorizes the intended upper-layer subject. The initial wire profile uses JWT/JWS and OAuth/OIDC-style claim names where they fit. Other encodings, including CWT/COSE or a signed manifest, are acceptable only if they preserve the same authority, freshness, canonicalization, and session-binding rules.

An Identity Grant contains the deployment-specific equivalent of:

- profile token type, `agtp_type=agtp.identity-grant` in the current JWT/JWS implementation;
- profile version, `agtp_version=1` in the current JWT/JWS implementation;
- issuer, `iss`;
- subject, `sub`;
- audience, `aud`;
- unique grant ID, `jti` for JWT/JWS or `cti` for CWT/COSE;
- issued-at time, `iat`;
- expiration time, `exp`;
- Agent confirmation key, `cnf.kid` in the initial JWT/JWS profile;
- optional explicitly authorized endpoint keys;
- service, tenant, deployment, or environment values when L3 is required;
- workload, process, or Agent identity when L4 is required;
- computation, task, thread, or delegation IDs when L5 is required;
- canonical intent, capability, or ontology references when those values are decision-sensitive;
- scopes, resources, or authorization details when L6 is required;
- issuer key ID or key version when needed for key rotation.

The Agent is not the authority for service, tenant, deployment, task, scope, resource, or capability values. Those values are accepted only when they appear in a grant verified under a locally trusted Manager or policy-authority key and then pass local policy comparison.

10. Session Binding Statement

A Session Binding Statement proves that the holder of the confirmation key named in the verified grant bound that exact grant to the accepted TLS or exported-authenticator session. It does not authorize identity or capability values.

A generic accepted endpoint key is not sufficient unless the verified grant or local attestation policy explicitly binds that key to the same Agent identity.

A Session Binding Statement contains the deployment-specific equivalent of:

- profile token type, `agtp_type=agtp.session-binding` in the current JWT/JWS implementation;
- profile version, `agtp_version=1` in the current JWT/JWS implementation;
- unique binding statement ID, `jti`;
- audience or relying-service ID, `aud`;
- issued-at time, `iat`;

- expiration time, exp;
- grant_hash;
- leaf_public_key_sha256;
- request_context_sha256;
- attestation_binder_sha256 when accepted attestation-to-channel evidence is present;
- one-shot nonce or unique binding value.

Binding means receiver-verifiable linkage to the accepted session and accepted evidence. The statement does not need to embed a TLS session identifier when the verifier can deterministically compare endpoint-key, exporter or request context, attestation-binder, nonce, and replay fields.

The grant hash is computed over an unambiguous byte string. For the initial encodings:

```
SHA-256("agtp.identity-grant.jwt.v1" || NUL || exact-signed-grant-bytes)
```

```
SHA-256("agtp.identity-grant.cwt.v1" || NUL || exact-signed-grant-bytes)
```

For JSON-based formats, the verifier SHOULD hash the exact signed bytes. It MUST NOT compute grant_hash by re-serializing JSON in the acceptance path. Canonical CBOR/COSE is acceptable when the encoding profile defines the canonical form.

When accepted attestation evidence includes an attestation-to-channel binder, attestation_binder_sha256 is REQUIRED in the Session Binding Statement and must match the accepted binder. When no such binder exists in the lower-layer session, the field is absent unless deployment policy requires it.

11. Verification model

The verifier performs authority checks, session-binding checks, replay checks, and policy comparison in that order. The final accepted assertion is built only from verified material.

1. Verify the Identity Grant under a trusted Manager or policy-authority key.
2. Check grant issuer, audience, profile version, token type, algorithm, key status, expiration, issued-at time, grant ID, and required scope or resource fields.
3. Verify the Session Binding Statement signature.
4. Check that the statement signer is the grant confirmation key or another endpoint key explicitly authorized by the verified grant or local policy.
5. Verify that grant_hash matches the exact verified Identity Grant bytes.
6. Verify that the statement audience matches the relying service or client.
7. Reject arbitrary accepted endpoint keys that are not explicitly bound to the same Agent identity by the verified grant or local attestation policy.
8. Compare the statement binding fields with the accepted TLS or exported-authenticator session.
9. Enforce replay policy for grant IDs, binding IDs, task-binding values, and nonces when one-shot use is required.
10. Construct the internal assertion only from the verified grant and binding statement.
11. Compare the assertion with local expected policy for every required layer.

identitypolicy.ValidateAssertion remains a comparator and binding freshness checker. It does not parse wire tokens, verify signatures, select keys, rotate keys, check revocation, or own replay storage. Those checks belong to the component that authenticates external identity material before it returns an observed assertion.

12. Policy model

Expected values and observed values stay separate.

- Expected values come from local policy, operator configuration, a trusted registry, a Manager, computation state, or a policy engine.
- Observed values come from a session-bound assertion built from authenticated grants, Session Binding Statements, attestation evidence, CoRIM metadata, EAT claims, agent manifests, authenticated request metadata, or authorization tokens.
- Verification compares observed values against expected values.
- Required layers fail closed when an expected value is missing, ambiguous, or only peer supplied.

A minimal policy object can be shaped as follows:

```
identity_policy:
  mode: "disabled"          # disabled | required
  set_mode: "contains_all" # contains_all | exact

  require:
    l3: false
    l4: false
    l5: false
    l6: false

  expected:
    service: ""
    tenant: ""
    deployment: ""
    environment: ""
    workload: ""
    agent: ""
    agent_public_key: ""
    computation_id: ""
    task_id: ""
    thread_id: ""
    delegation_id: ""
    intent_ref: ""
    capability_ref: ""
    ontology_id: ""
    scopes: []
    resources: []
    authorization_details: []
```

After external identity material has been authenticated, the verifier can build an internal assertion:

```
identity_assertion:
  issuer: "manager-or-policy-engine"
```

```

values:
  service: "payment-agent"
  tenant: "tenant-a"
  deployment: "prod"
  environment: "asia-northeast1"
  workload: "settlement-worker"
  agent: "agent-a"
  agent_public_key: "sha256:..."
  computation_id: "cmp-..."
  task_id: "task-..."
  thread_id: "thread-..."
  delegation_id: "delegation-..."
  intent_ref: "intent:monthly-settlement"
  capability_ref: "capability:orders-read"
  ontology_id: "ontology:payments:v1"
  scopes:
    - "orders:read"
  resources:
    - "orders"
  authorization_details:
    - "purpose:monthly-settlement"

binding:
  leaf_public_key_sha256: "..."
  request_context_sha256: "..."
  attestation_binder_sha256: "..."
  nonce: "..."
  issued_at: "..."
  expires_at: "..."

```

This assertion is not a wire format. It is a local representation used after the caller has authenticated external identity material. Assertion.Issuer is informational unless the observed-identity implementation has already verified the corresponding issuer and trust anchor.

13. L3 through L6 policy fields

Layer	Required question	Common expected values
L3	Is this the intended service, tenant, deployment, or environment?	service, tenant, deployment, environment, region, CoRIM or EAT deployment claims

Layer	Required question	Common expected values
L4	Is this the intended workload, process, or Agent?	workload ID, Agent ID, process or binary hash, config or policy hash, Agent public key, routing target
L5	Is this the intended task, thread, context, or delegation?	computation ID, task ID, thread ID, request context, delegation ID, callback binding, one-shot task state
L6	Is the action authorized for this peer and context?	scope, resource, authorization details, capability token, policy decision, user consent, tool policy

OIDC and OAuth-style names are useful analogies:

Layer	OAuth/OIDC-style analogue
L3	issuer, audience, tenant, hosted domain, deployment claim, environment claim
L4	subject, client ID, workload identity, actor claim, confirmation key
L5	nonce, state, transaction ID, request object, delegation token
L6	scope, resource indicator, authorization details, policy decision, consent

These analogies are not normative requirements. A deployment can source the same expected values from configuration, a registry, CoRIM metadata, an EAT claim, an agent manifest, or an application policy engine.

14. Canonical semantic references

Some L5 and L6 inputs are semantic references rather than raw labels:

- `ontologyId`: stable identifier for the vocabulary or policy namespace;
- `intentRef`: stable identifier for the intended task, intent, or action family;
- `capabilityRef`: stable identifier for an authorization capability or resource-action class.

These values are profile reference identifiers. They are not natural-language prompts, display names, or peer-selected aliases.

Decision-sensitive semantic values **MUST** be canonical before they appear in an Identity Grant, Session Binding Statement, local expected policy, or authorization decision. The relying client, Manager, or local policy authority resolves aliases before issuance or local policy comparison.

Receivers **MUST** validate canonical identifiers deterministically against receiver-side policy, task state, trusted registry data, signed profile data, or the underlying interaction/security protocol.

Receivers **MUST NOT** normalize peer-provided decision-sensitive values during the final acceptance path. They **MUST NOT** turn a peer-provided alias, display label, natural-language phrase, case variant, URI variant, or model interpretation into an accepted canonical identifier.

Alias handling is part of local policy:

- an alias resolves to exactly one normalized reference at the relevant registry or policy version;
- a missing alias, unsupported registry version, or ambiguous alias fails closed;
- case folding, Unicode normalization, URI normalization, or prefix expansion is allowed only when the referenced identifier scheme defines it;
- otherwise, normalized references are compared as exact strings;
- set-like fields, such as scopes, resources, and authorization details, use the configured set mode after normalization.

Free-form natural-language intent, context, or capability text MAY be carried as descriptive metadata. It MUST NOT be decision-authoritative profile data.

15. Production profile

Production identity policy has two modes: disabled or required.

In disabled mode, the transport does not enforce L3 through L6 identity policy. L0 through L2 lower-layer checks still apply according to the configured TLS and attestation profile.

In required mode, the client fails closed when any required policy input is missing, untrusted, expired, replayed, non-canonical, unsupported, revoked, or inconsistent with the accepted TLS session. Peer-provided metadata is never used as expected policy.

Area	Required-mode behavior
Manager trust	Manager or policy-authority verification keys are configured locally by kid, algorithm, issuer, audience, key use, and public key, or loaded through an equivalent fail-closed key source.
Algorithm safety	The token alg must match local key policy. none is forbidden. Algorithm confusion is a hard failure.
Key separation	Manager signing keys and Agent confirmation keys are separate trust domains and MUST NOT be accepted interchangeably.
Grant lifetime	Identity Grants are short-lived and carry iat, exp, and a unique jti or cti.
Grant revocation	Deployments that require early invalidation reject revoked grant IDs and disabled Manager-key IDs before expiration.
Unknown keys	Unknown, disabled, retired, stale, wrong-use, or mismatched key IDs fail closed.
Session-binding signer	Session Binding Statements are signed by the confirmation key named in the verified grant, or by another key explicitly authorized by that grant or local policy.
Session-binding fields	Session Binding Statements carry accepted endpoint-key hash, request/exporter-context hash, one-shot nonce, and attestation-binder hash when attestation binding is present.

Area	Required-mode behavior
Replay	JWT/CWT identity acceptance requires replay protection for binding nonces or one-shot IDs. Multi-instance production deployments use shared atomic insert-with-expiry semantics, such as SET NX EX.
Local policy	Required L3 through L6 values are compared against local expected policy. Missing expected values fail closed.
Error handling	Failure to check keys, revocation, session binding, replay, canonicalization, or local identity policy is an authentication failure, not a warning.

The initial mandatory JWT/JWS claim set is:

Token	Mandatory claims
Identity Grant	agtp_type, agtp_version, iss, aud, sub, jti, iat, exp, cnf.kid
Session Binding Statement	agtp_type, agtp_version, aud, jti, iat, exp, grant_hash, leaf_public_key_sha256, request_context_sha256, nonce; also attestation_binder_sha256 when attestation binding is present

When identity is accepted from JWT/JWS material, the Session Binding Statement JWT is REQUIRED. The Identity Grant JWT authenticates authority and semantics; it does not prove that the authorized Agent key is present on the current TLS or exported-authenticator session. The verifier accepts identity only after the Session Binding Statement JWT is verified, linked to the exact signed Identity Grant JWT, compared with the accepted session binding, checked for freshness and replay, and compared with local expected policy.

CWT/COSE is a compact binary encoding for the same model. The grant issuer must be trusted locally, the confirmation key must be named by the verified grant, the COSE kid used for key lookup must be protected, and the binding statement must bind the exact grant to the accepted session.

16. Freshness, replay, key rotation, and revocation

JWT/JWS and CWT/COSE adapters verify signatures against caller-provided key lookup policy. They do not define how Manager keys are rotated, how old keys are retired, how grants are revoked before expiration, or where replay state is stored. Deployments own those decisions.

Deployments using this profile define:

- a trusted issuer namespace for Manager or policy-authority keys;
- key identifiers, key-use rules, and key-version rules;
- overlap windows for key rotation;
- revocation sources for grants, Manager keys, or Agent binding keys when early invalidation is required;
- maximum grant and session-binding lifetimes;
- clock-skew tolerance;
- replay-cache key shape;

- replay-cache retention at least as long as the accepted session-binding lifetime.

Revocation has three separate targets:

- grant revocation rejects a specific Identity Grant by JWT `jti` or CWT `cti`;
- Manager-key revocation rejects grants signed by a disabled issuer key or `kid`;
- Agent binding-key revocation rejects Session Binding Statements signed by a compromised or retired confirmation key.

Unknown, revoked, stale, wrong-use, or ambiguous keys fail closed. A deployment that relies on JWKS or another remote key set must define freshness, cache lifetime, pinning or trust anchors, and failure behavior for that key source.

Replay state is marked only after signature validation, session-binding comparison, freshness checks, and local policy comparison succeed. Failed attempts may be logged or rate-limited, but they do not consume a one-shot nonce unless deployment policy deliberately chooses that anti-probing behavior.

17. Application response-cache semantics

Response caching and security-state caching are different things.

Response caching stores endpoint results for reuse. Replay caches store one-shot freshness state. Verified Identity Grants, Session Binding Statements, attestation evidence, authorization decisions, and prior verification results **MUST NOT** be cached as acceptance evidence for a later session or request.

Application response-cache controls, if used, follow RFC 9111's split between shared and private caches, freshness, and response controls. The default for profile-sensitive endpoints is `no-store`.

Directive	Meaning
<code>no-store</code>	A cache MUST NOT store the response. This is REQUIRED for identity tokens, Session Binding Statements, attestation evidence, capability grants, authorization decisions, and responses containing session-bound or caller-sensitive data.
<code>public</code>	A shared cache MAY store the response only when the response is invariant across Agent ID, principal, tenant, authority scope, policy grant, mTLS certificate identity, and session context.
<code>private</code>	A shared cache MUST NOT store the response. An agent-local or principal-local cache MAY store it only inside the relevant identity or policy partition.
<code>max-age</code>	Response freshness lifetime. It does not extend token, grant, binding, revocation, attestation, or local policy lifetimes.
<code>vary</code>	Identity, policy, or request fields that partition a cached response when the response can differ by caller or authority context.

`public` is a strong assertion. A public response must not depend on caller identity, scope, principal, tenant, policy grant, mTLS certificate, attestation result, session binding, task state, or any other verifier-local policy input.

Private cache hits do not bypass current security checks. A verifier still authenticates the current Identity Grant, verifies the current Session Binding Statement, checks replay state, and compares local policy before using a cached response. Cache hit is a performance result, not an authorization result.

18. Deployment topologies

18.1 Direct-Agent mode

In direct-Agent mode, the Agent terminates the TLS session and performs the hardware-aware profile checks. The Agent signs the Session Binding Statement with the confirmation key named by the verified Identity Grant, or with an endpoint key explicitly authorized by the grant or local policy.

18.2 Gateway-routed mode

In gateway-routed mode, the gateway terminates the TLS session and performs the hardware-aware profile checks. The gateway is the live TLS endpoint. Gateway session binding proves the gateway endpoint, not the final Agent process.

The deployment must also authenticate the gateway-to-Agent route before treating the intended Agent as accepted. That can be modeled with gateway ID, gateway public key, allowed route, and Agent route fields in the Identity Grant, or with a separate gateway routing assertion.

A gateway routing assertion is not an Agent authority statement. It authenticates the gateway's routing decision. The relying party still needs an Identity Grant for the intended Agent and, when required, an Agent-side holder-of-key proof for the gateway-to-Agent hop.

19. Implementation hooks

The production client hook is:

- `atls.ClientConfig.IdentityPolicy` for local expected values;
- `atls.ClientConfig.ObservedIdentity` for a session-bound observed assertion extracted from a trusted source.

The hook runs after exported-authenticator and attestation validation, but before the accepted TLS connection is returned to the caller.

The client computes expected session binding from the accepted exported authenticator: the leaf public key, the certificate request context, and the attestation binder when attestation is present. The observed assertion must carry matching binding values and a non-expired `expires_at` value before its identity fields are compared with local policy.

The profile gives the client two defenses:

- relay defense: the observed assertion is tied to this accepted TLS session at L2;
- diversion defense: the session-bound observed identity matches locally expected deployment, Agent, task, and authorization values at L3 through L6.

20. Validation algorithm

For each layer required by policy:

1. Load the local expected value.

2. Reject if the expected value is missing, ambiguous, non-canonical, or only peer supplied.
3. Extract the observed session-bound assertion from an authenticated source.
4. Reject if the assertion is not bound to the accepted TLS session.
5. Reject if the assertion is expired or missing freshness metadata.
6. Extract the observed value for the required layer.
7. Reject if the observed value is missing or non-canonical.
8. Compare the observed value with the expected value.
9. Reject on mismatch.
10. Continue to the next required layer.

For set-like values such as scopes, resources, or authorization details, the observed set must satisfy local policy. The default `contains_all` mode accepts an observed set only when it contains every locally required value. The stricter `exact` mode also rejects extra observed values. A peer-provided scope list is not enough by itself.

21. Error handling and diagnostics

Validation failures preserve the layer, field, and error class. Callers can distinguish local policy configuration errors from missing peer claims, mismatched values, stale keys, replay, unsupported profile versions, and binding mismatches.

Aggregated validation errors remain inspectable by layer and field so callers can fail closed while still reporting actionable diagnostics.

The validator rejects unsafe identity-policy values, including invalid UTF-8, control characters such as CRLF, and HTML delimiter characters. It limits individual values to 1024 bytes and set-like fields to 128 values. Validation errors report only layer, field, and error class; they do not echo raw peer values. HTTP or HTML presentation layers still need normal output escaping and CSRF protections.

22. Minimal implementation path

A small implementation can be staged without changing the lower-layer TLS or post-handshake attestation wire protocol:

1. Define a local policy input structure for L3 through L6 expected values.
2. Define extraction points for observed values.
3. Authenticate the observed identity material before constructing an internal assertion.
4. Add fail-closed validators for exact-match string fields.
5. Add set-containment or exact-set validation for scopes, resources, and authorization details.
6. Add replay-cache support for one-shot binding nonces or IDs.
7. Wire validation before treating an accepted peer as the intended deployment, Agent, task, or authorized actor.

The lower-layer verifier can remain focused on L0 through L2. L3 through L6 enforcement belongs at the layer that has deployment policy, Agent metadata, computation state, and authorization decisions.

23. Implementation status

The reusable validator lives in `pkg/atls/identitypolicy`. It implements the expected-versus-observed comparison model and session-bound assertion validation described in this profile. The client transport calls it when `atls.ClientConfig.IdentityPolicy` is enabled.

The same package provides `identitypolicy.NewAssertionFromSessionBinding` for the post-authentication step. It builds an internal Assertion from an already authenticated Identity Grant and an already-verified Session Binding Statement. It does not parse wire tokens or verify signatures. It checks that the statement is tied to the grant, audience, allowed confirmation key, and minimum session-binding fields before policy comparison. `identitypolicy.SessionBindingOptions` can attach a replay cache for one-shot binding nonces.

The client config supports two integration styles. Callers can provide a custom `ObservedIdentity` callback, or they can pass a verified Identity Grant, a verified Session Binding Statement, and an optional replay cache directly on `atls.ClientConfig`. In both cases, the transport validates the resulting assertion against the accepted TLS session before returning the connection. For the direct grant/statement path, replay-cache marking happens only after session-binding and local policy comparison succeed.

`pkg/agtp` provides concrete JWT/JWS and CWT/COSE wire-token adapters. The JWT/JWS adapter verifies Identity Grants and Session Binding Statements using locally configured issuer, audience, signing methods, and key lookup policy. It does not choose trusted Manager keys, rotate keys, perform revocation, define deployment policy, or own distributed replay storage.

The initial JWT/JWS adapter treats `cnf.kid` as the authorized session-binding signer key. The Session Binding Statement signer is taken from the protected JWS kid header and later checked by `identitypolicy` against the verified grant. Thumbprint-based confirmation, such as `cnf.jkt`, can be added when a deployment defines how key thumbprints map to local verification keys.

For callers that want one fail-closed acceptance gate, `pkg/agtp` exposes `VerifySessionIdentityJWT`. That helper verifies the Manager-signed Identity Grant, verifies the Session Binding Statement, checks binding-signer authorization, compares the resulting assertion with local expected `identitypolicy.Policy` values and the accepted TLS session binding, and only then marks the binding nonce in the replay cache.

The implemented production profile covers:

- trusted Manager or policy-authority key lookup for Identity Grants through `JWTVerifyOptions`;
- issuer, audience, signing-method, kid, expiration, issued-at, token type, profile version, and jti checks;
- confirmation-key binding through Identity Grant `cnf.kid`;
- Session Binding Statement signer authorization against the verified grant;
- comparison with the accepted TLS endpoint key, request context, and optional attestation binder;
- local `identitypolicy.Policy` comparison for required L3 through L6 values;
- replay-cache enforcement before the observed identity is accepted.

Deployment still chooses trusted keys, expected policy values, revocation data, and distributed replay storage. Those sources can be Manager configuration, Agent metadata, computation state, an authorization policy engine, or a fail-closed registry integration. They must not be raw peer-controlled metadata.

Appendix A. Identity Grant JWT claim map

The initial JWT/JWS profile uses two signed JWTs for different authority questions:

- Identity Grant JWT: a Manager or policy authority signs upper-layer identity, task, and authorization values that the peer is allowed to claim.
- Session Binding Statement JWT: the Agent confirmation key signs a holder-of-key statement binding the verified grant to the accepted TLS session.

The Session Binding Statement JWT does not authorize service, tenant, task, scope, resource, or capability values. Those semantic values belong in the Identity Grant JWT and are accepted only after local policy comparison.

Identity Grant JWT field	Source or form	Purpose	Layer
JWS protected header kid	Manager or policy-authority signing key ID	Select the local verification key for the signed grant. The value is only a key-selection hint.	prerequisite
JWS protected header alg	locally allowed signing method	Prevent algorithm confusion. none is forbidden.	prerequisite
agtp_type	agtp.identity-grant	Distinguish the grant token from other repository profile tokens.	profile guard
agtp_version	1	Reject unsupported wire-profile versions before interpreting claims.	profile guard
iss	trusted Manager or policy authority	Identify the authority that issued the grant.	prerequisite
aud	relying service or application profile audience	Ensure the grant was issued for this verifier or application profile.	prerequisite
sub	upper-layer subject, usually the Agent when agent is absent	Provide the subject authorized by the grant.	L4
jti	unique grant ID	Support grant uniqueness, revocation, diagnostics, and replay-sensitive handling.	freshness
iat	issued-at time	Detect future-issued or malformed grants and support freshness decisions.	freshness
exp	expiration time	Bound the lifetime of the authority statement.	freshness
cnf.kid	Agent confirmation key ID	Name the key allowed to sign the Session Binding Statement for this grant.	L4 / L2 prerequisite
authorized_endpoint_keys	optional endpoint key IDs	Allow explicitly named endpoint keys when the deployment uses that binding model.	L4 / L2 prerequisite
service	application service name or ID	Bind the grant to the intended service.	L3
tenant	tenant or account ID	Bind the grant to the intended tenant.	L3
deployment	deployment, region, cluster, or environment slice	Bind the grant to the intended deployment target.	L3
environment	production, staging, development, or similar	Bind the grant to the intended environment class.	L3
workload	workload, process, or component ID	Bind the grant to the intended workload.	L4

Identity Grant JWT field	Source or form	Purpose	Layer
agent	Agent identity	Bind the grant to the intended Agent rather than any peer on the same platform.	L4
agent_public_key	stable Agent public-key reference or fingerprint	Bind the Agent identity to a specific key when required by policy.	L4
computation_id	computation or job ID	Bind the grant to a computation instance.	L5
task_id	task ID	Bind the grant to the intended task.	L5
thread_id	thread, conversation, or workflow ID	Bind the grant to the intended thread or workflow context.	L5
delegation_id	delegation token or delegation record ID	Bind the grant to a specific delegation chain or authorization handoff.	L5
intent_ref	stable intent reference	Bind the grant to a canonical task or intent family.	L5
capability_ref	stable capability reference	Bind the grant to a canonical capability or resource-action class.	L6
ontology_id	stable policy vocabulary or namespace ID	Identify the vocabulary used to interpret semantic references.	L6
scope / scopes	OAuth-style scope values	State coarse authorization strings for local policy comparison.	L6
resource / resources	resource indicators or resource URIs	State the resources the grant may be used against.	L6
authorization_details	profile-specific authorization detail strings	Carry fine-grained authorization constraints.	L6

Appendix B. Session Binding Statement JWT claim map

The Session Binding Statement JWT carries session facts, not semantic authority. A verifier that accepts JWT/JWS identity material **MUST** verify this JWT under the Agent confirmation key authorized by the verified Identity Grant and **MUST** reject missing, mismatched, expired, replayed, non-canonical, or unsupported binding material.

Session Binding Statement JWT field	Source or form	Purpose	Layer
JWS protected header kid	Agent confirmation or explicitly authorized endpoint key ID	Select the candidate binding-signature key. The key is accepted only if authorized by the verified grant or local policy.	L2/L4 prerequisite

Session Binding Statement JWT field	Source or form	Purpose	Layer
JWS protected header alg	locally allowed signing method	Prevent algorithm confusion. none is forbidden.	prerequisite
agtp_type	agtp.session-binding	Distinguish the binding token from grants, envelopes, or unrelated JWTs.	profile guard
agtp_version	1	Reject unsupported wire-profile versions before interpreting binding fields.	profile guard
aud	relying service or application profile audience	Ensure the statement was issued for this verifier or profile.	L2 prerequisite
jti	unique binding statement ID	Support diagnostics and one-shot replay handling.	freshness
iat	issued-at time	Detect future-issued or malformed binding statements.	freshness
exp	expiration time	Bound the lifetime of the session proof. It should not exceed the grant lifetime.	freshness
grant_hash	domain-separated hash of exact signed Identity Grant bytes	Bind the statement to the exact Manager-signed grant that was verified.	L2
leaf_public_key_sha256	hash of accepted TLS or exported-authenticator endpoint public key	Bind the statement to the accepted endpoint key.	L2
request_context_sha256	hash of accepted request/exported-authenticator context	Bind the statement to the accepted request context.	L2
attestation_binder_sha256	hash of accepted attestation binder, when present	Bind the statement to accepted attestation-to-channel evidence.	L2
nonce	one-shot nonce or unique binding value	Prevent replay of an otherwise valid binding statement.	freshness

The receiver compares these fields as strict, already-canonical values. It does not repair aliases, reserialize JSON to compute `grant_hash`, infer missing binding fields from peer metadata, or treat a valid Identity Grant JWT as session-bound without the Session Binding Statement JWT.

Appendix C. Authority separation and envelope packaging

A deployment may package the Manager-signed Identity Grant inside, or alongside, the Agent-signed Session Binding Statement so an implementation passes around one envelope. Packaging does not merge authorities. The verifier still validates the Manager or policy-authority signature on the grant,

validates the Agent confirmation-key signature on the session binding, checks that the binding names the exact verified grant, and compares the resulting assertion with local policy.

Moving semantic grant fields directly into an Agent-signed Session Binding Statement would make the Agent the authority for service, tenant, task, scope, resource, and capability claims. That changes the trust model and is unsafe for profiles that need Manager- or policy-authority-issued authorization.

A single Manager-signed JWT that also includes session-binding values is possible only when the Manager participates in, or is given trustworthy evidence of, the per-connection TLS binding. That design adds an online authority dependency and moves replay and freshness responsibilities to the Manager path.

The two-JWT design adds one additional JWS verification plus a domain-separated hash over the exact compact Identity Grant JWT bytes. This is usually small compared with TLS setup and attestation verification. Verifiers MUST NOT cache a verified Identity Grant as acceptance evidence across sessions. Internal optimization must still re-check signature, expiration, revocation state, profile version, issuer, audience, grant hash, fresh Session Binding Statement, replay state, and local policy for each accepted session.

The reusable pkg/agtp JWT/JWS adapter also supports a single-envelope verification path. In that profile, the outer envelope is signed by the Agent binding key and carries the exact inner Manager-signed Identity Grant JWT plus the session-binding fields. The verifier still authenticates the inner Manager grant, authenticates the outer Agent binding statement, compares the outer grant_hash with the exact inner grant bytes, enforces replay and freshness, and compares the resulting assertion with local policy. Agent-signed semantic claims in the outer envelope are not Manager authorization. The current runtime client configuration remains wired to the two-token JWT/JWS profile unless a caller explicitly uses the envelope verifier.

Appendix D. Non-normative trust-flow example

sequenceDiagram

participant M as Bob Manager

participant B as Bob Agent

participant A as Alice

participant X as Malware Agent on Alice machine

M->>B: Identity Grant signed by Manager key

Note over M,B: Grant authorizes Bob Agent key

A->>B: Start TLS or hardware-aware TLS session

B->>A: Identity Grant

B->>A: Session Binding Statement signed by Bob Agent key

X-->>A: Injects peer-provided identity metadata

X-->>A: Replays another grant or binding statement

A->>A: Verify grant with trusted Bob Manager public key

A->>A: Verify session binding with Bob Agent key from grant

A->>A: Check expected service, Agent, task, and capability

A->>A: Reject metadata not signed or not bound to this session

Bob Manager issues an authorization grant for Bob Agent. Bob Agent communicates with Alice and binds the Manager grant to the current TLS or hardware-aware TLS session. Malware on Alice's machine can try to inject peer-provided metadata or reuse an old grant or binding statement, but it cannot create a valid Manager-signed grant or Bob-Agent-key-signed session binding for the current accepted session.

Alice accepts only the Manager-signed grant and the Agent-signed session binding after local expected service, Agent, task, and capability checks pass. This does not protect a fully compromised Alice process or local policy store.